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AHEAD Online

Object Oriented Programming Using Java

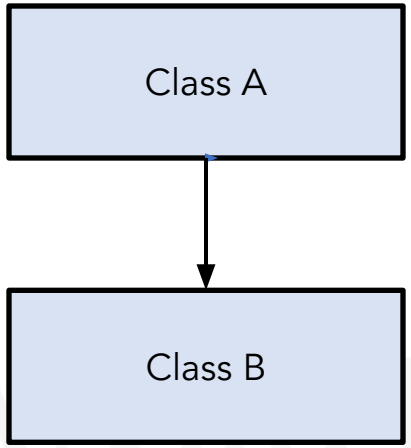
Objectives

- Types of Inheritance.

Types of Inheritance

- Single Inheritance
- Multilevel Inheritance
- Hierarchical Inheritance
- Hybrid Inheritance.

Single Inheritance



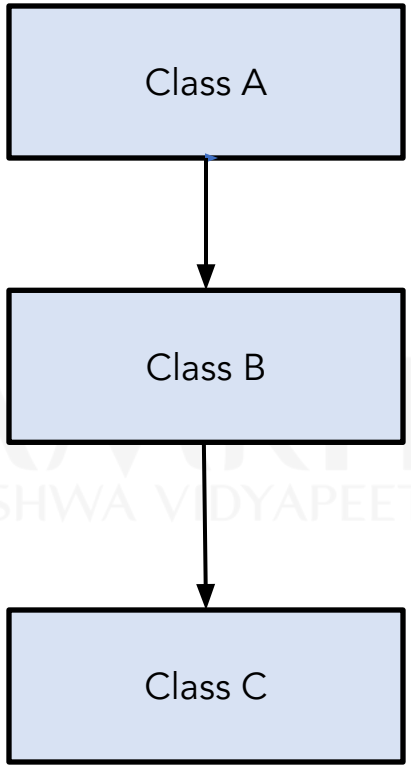
- A is the base class.
- B is the derives class.

Single Inheritance

```
class A
{
}
class B extends A
{
}
```

- Class B can access all the members of class A except private members.

Multilevel Inheritance

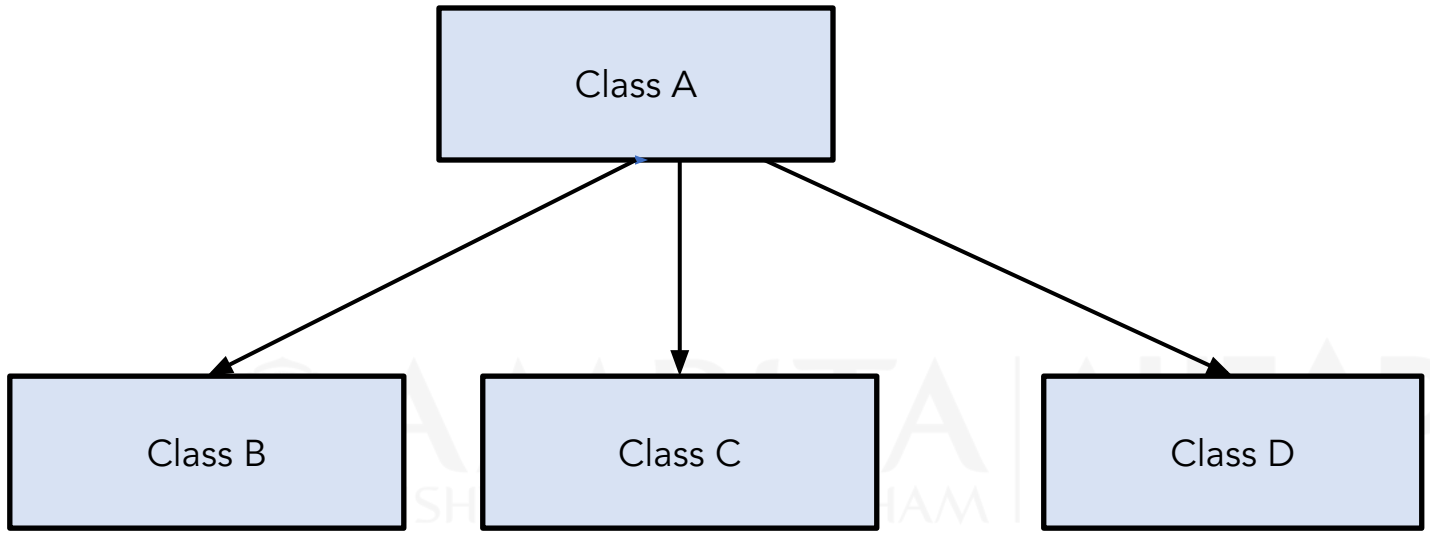


Multilevel Inheritance

```
class A
{
}
class B extends A
{
}
Class C extends B
```

- Class C is the subclass of B which is the subclass of A.

Hierarchical Inheritance

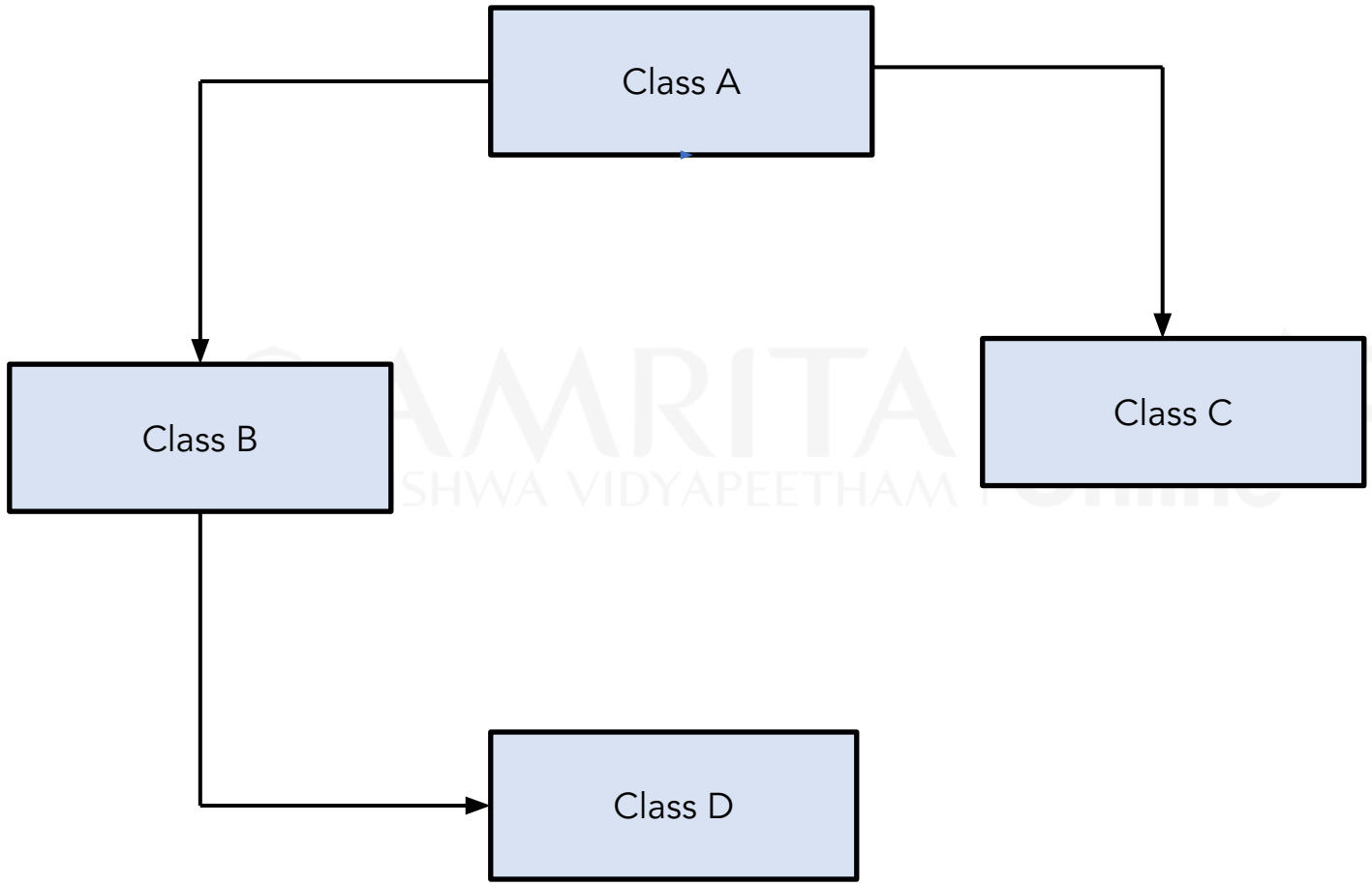


Hierarchical Inheritance

```
class A
{
}
class B extends A
{
}
Class C extends A
{
}
Class D extends A
{
}
```

Classes B, C and D inherit the class A

Hybrid Inheritance



Hybrid Inheritance

```
class A
{
}
class B extends A
{
}
Class C extends A
{
}
Class D extends B
{
}
```

A combination of more than one type of inheritance(Hierarchical and single inheritance) Classes B, C extends A and D inherit the class B.

Takeaways

- Types of Inheritance.

Reference

[1] Herbert Schildt, *"Java: The Complete Reference, Eleventh Edition"*, *Oracle 2018*